



“Team members”

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Project idea

Project Happiness: Unlocking Global Well-being

We are embarking on an exciting data journey using the World Happiness Ranking dataset from Kaggle. This dataset connects nations with critical socio-economic and state-level factors to explain their reported Happiness Score.

Our goal is three-fold:

- Data Harmony: Clean and prepare the raw data (addressing missing values and errors) to ensure accuracy and consistency.
- Visual Joy: Design an insightful and aesthetically pleasing Power BI Dashboard that clearly visualizes global happiness trends and their underlying drivers.
- The Happy Story: Create a PowerPoint Report that narrates our findings, explains the cleaning process, details the dashboard's purpose, and presents the key conclusions derived from the analysis.

Let's transform this data into actionable insights about global well-being!



Dataset description

Dataset Overview:

This dataset contains happiness and well-being metrics for countries worldwide across the years 2015 and 2016. The data captures multiple dimensions that contribute to national happiness, providing a comprehensive view of global well-being patterns and regional differences.

The dataset structure includes:

- 314 total records (157 per year)
- 7 core analytical columns plus region and year identifiers
- Mixed data types: categorical (countries/regions) and numerical scores
- Available Data Columns

The dataset focuses on these key happiness indicators:

- 1) Country – Nation name
- 2) Region – Geographic/cultural grouping
- 3) Happiness Rank – Global ranking position
- 4) Happiness Score – Overall happiness index (0-10 scale)
- 5) Economy (GDP per Capita) – Economic contribution factor
- 6) Family – Social support and family connections factor
- 7) Health (Life Expectancy) – Life expectancy and healthcare factor
- 8) Freedom – Perceived freedom in life choices
- 9) Year – Data collection year (2015 or 2016)

1. Zero Values in Continuous Metrics

Issue: Several countries show 0.00 values in Health, Family, and Freedom columns

Example: Central African Republic shows 0.000 in Family (2015), which is unrealistic.

Solution: Handle Zero/Unrealistic Values, Replace zeros in continuous columns (Health, Family, Freedom, Economy) with NaN.

Reason: Zeros likely indicate missing data rather than true zero contributions

2. Extreme Outliers

Issue: Very low scores in Economy or Health for some countries (e.g., near-zero values).

Solution: Investigate whether these are data errors or true reflections (e.g., conflict zones, low-income nations).

3. Missing Data Handling

Issue: Some columns had over 90% missing values, making them unusable for analysis.

Solution: Those columns were removed entirely from the dataset to ensure reliability and avoid misleading analysis.





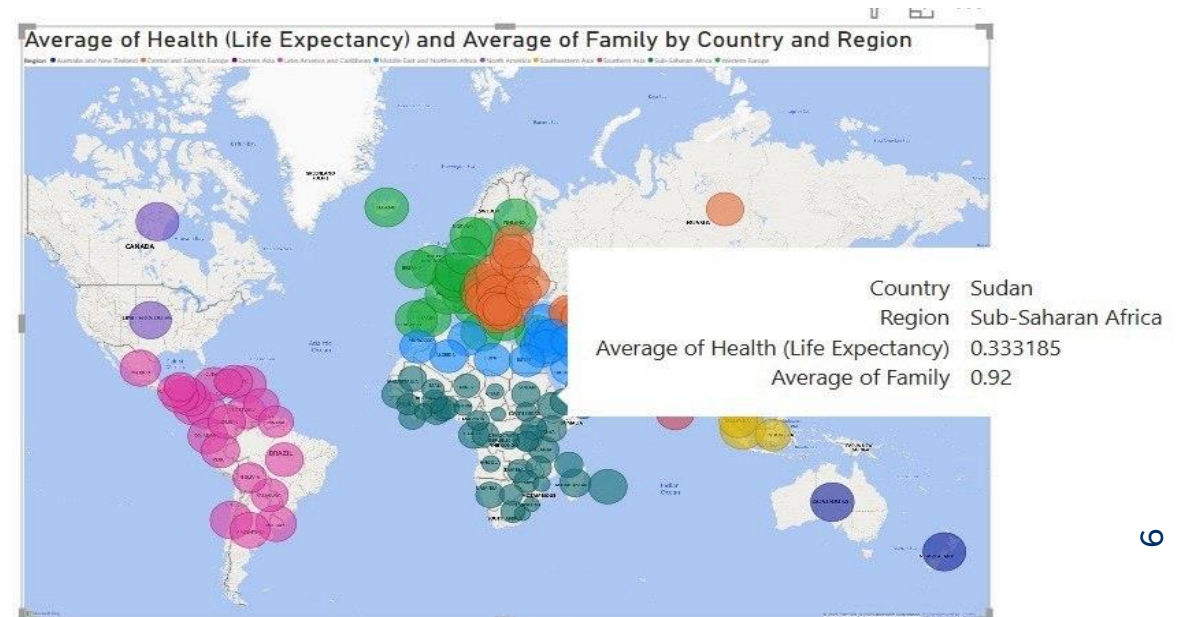
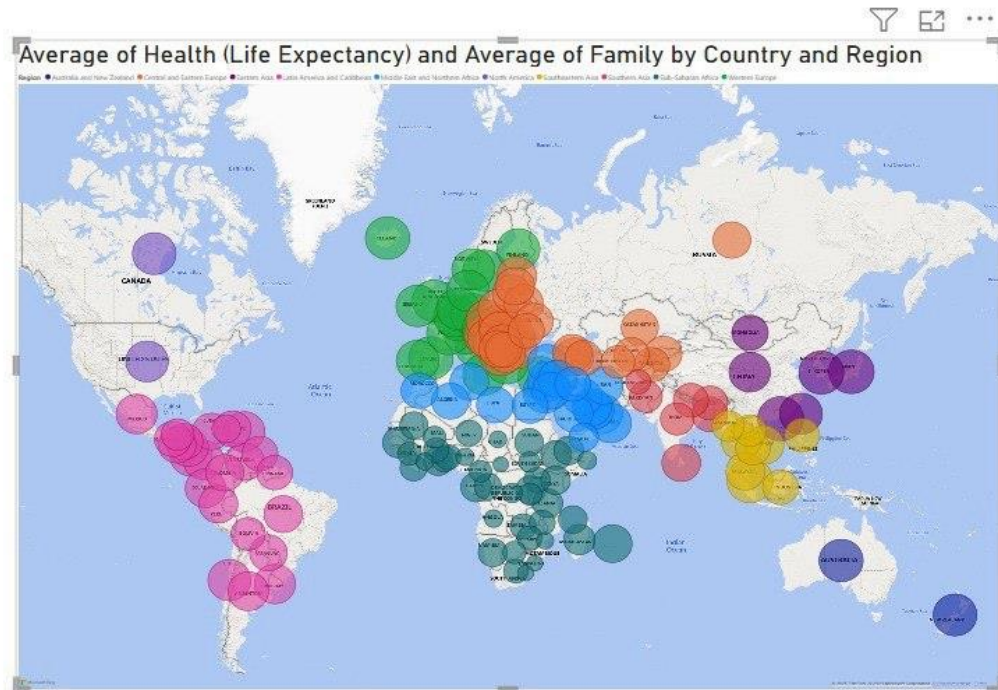
Data Visualization Descriptions

1. Country & Region Health-Family Analysis

Chart Type: Detail Table / Summary Card

Purpose: Shows the relationship between average Health (Life Expectancy) and Family (social support) metrics for specific countries within their regions.

Insight Example: Sudan (Sub-Saharan Africa) shows relatively low health scores (0.33) but moderate family support scores (0.92), indicating a potential disconnect between healthcare and social structures in that region.





2. Top 10 Countries by Happiness Score

Chart Type: Bar Chart (Horizontal)

Purpose: Ranks the 10 happiest countries based on their average happiness scores, providing a clear hierarchy of well-being.

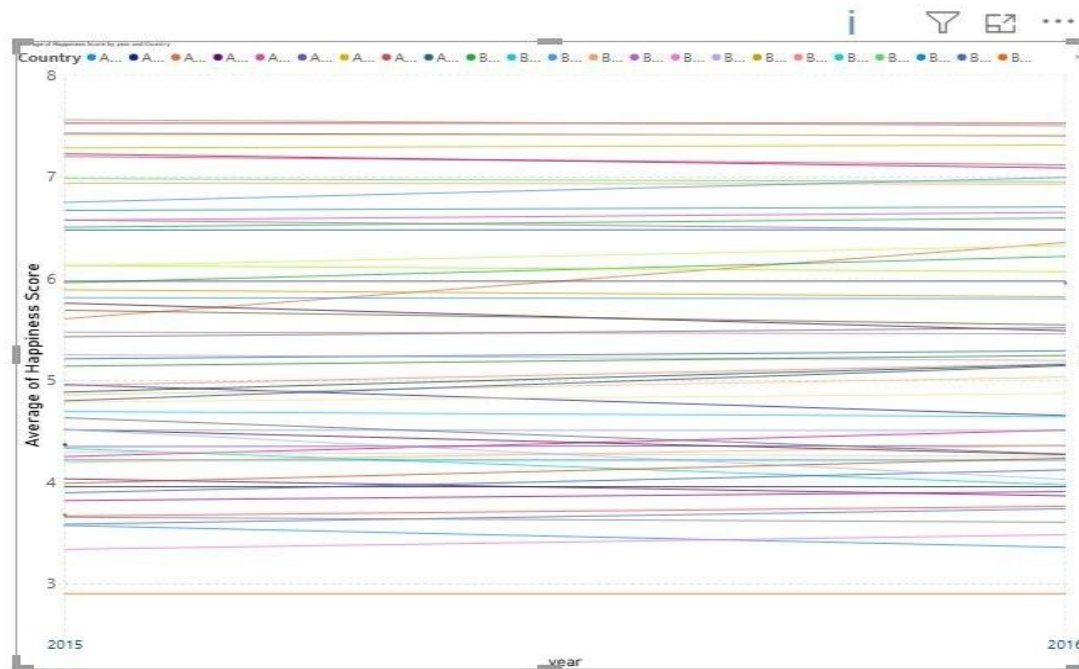
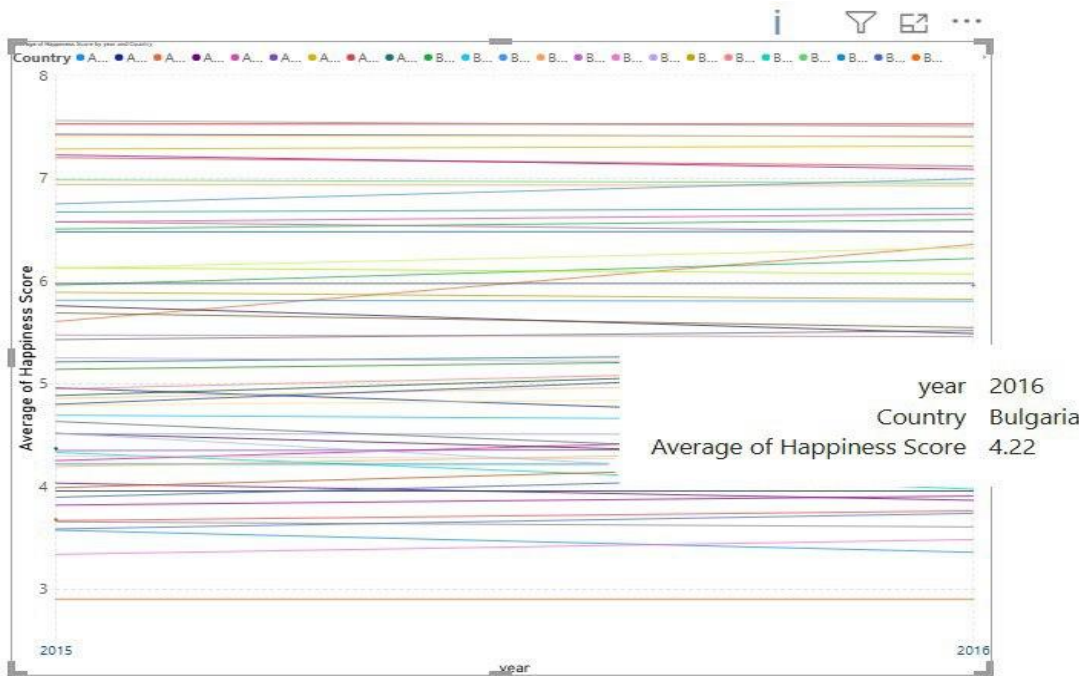
Key Insight: Switzerland leads the ranking (~7.60), followed closely by Iceland and Denmark, with all top 10 countries maintaining scores above 7.25. Shows consistent happiness leadership among Western European and Anglo-Saxon nations.

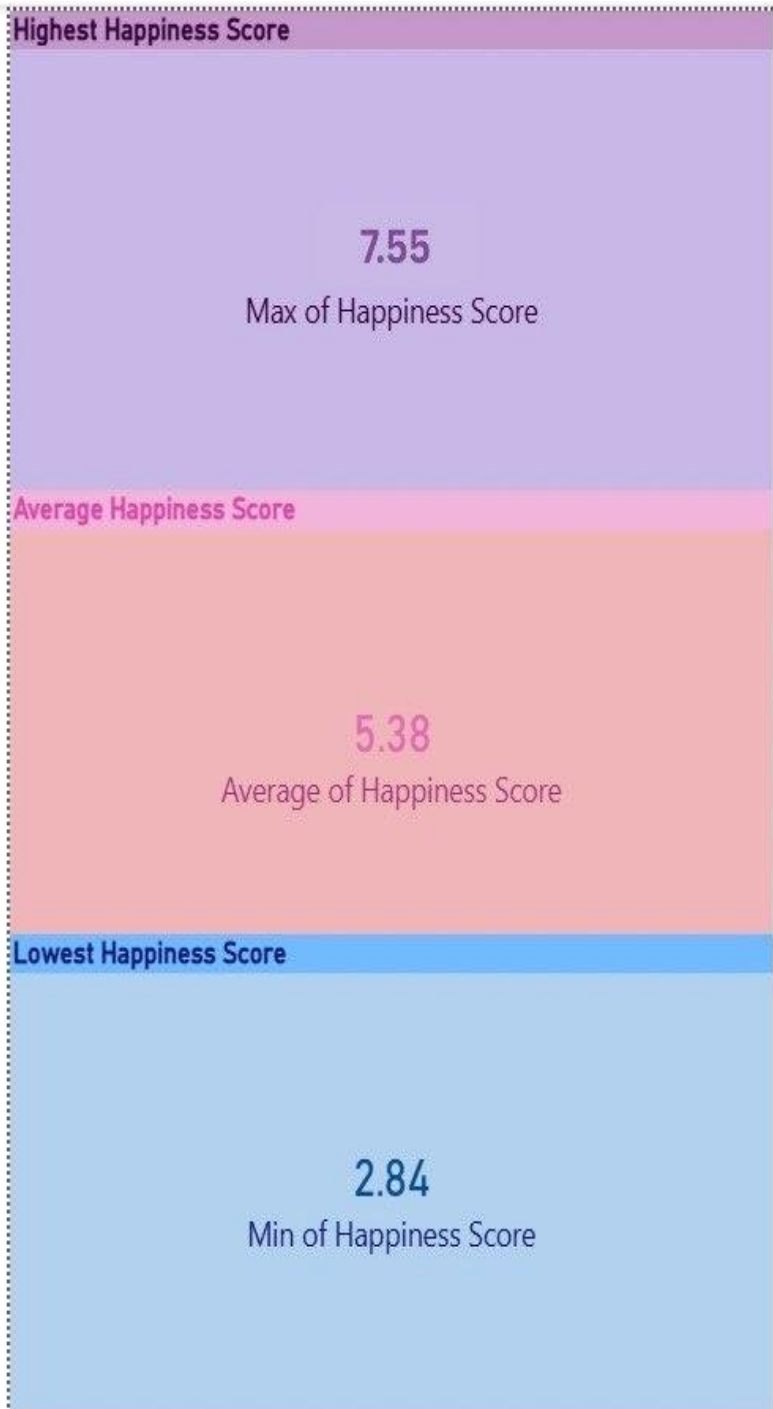
3. Year-over-Year Happiness Comparison

Chart Type: Comparison Card / Dual Value Display

Purpose: Compares happiness scores for the same country (Bulgaria) across two years (2015 vs 2016).

Insight: Bulgaria's happiness score remained relatively stable at approximately 4.22 between 2015 and 2016, suggesting little change in overall well-being during that period.





4. Happiness Score Distribution Summary

Chart Type: Statistical Summary Card

Purpose: Displays key statistical metrics for the happiness score variable across the entire dataset.

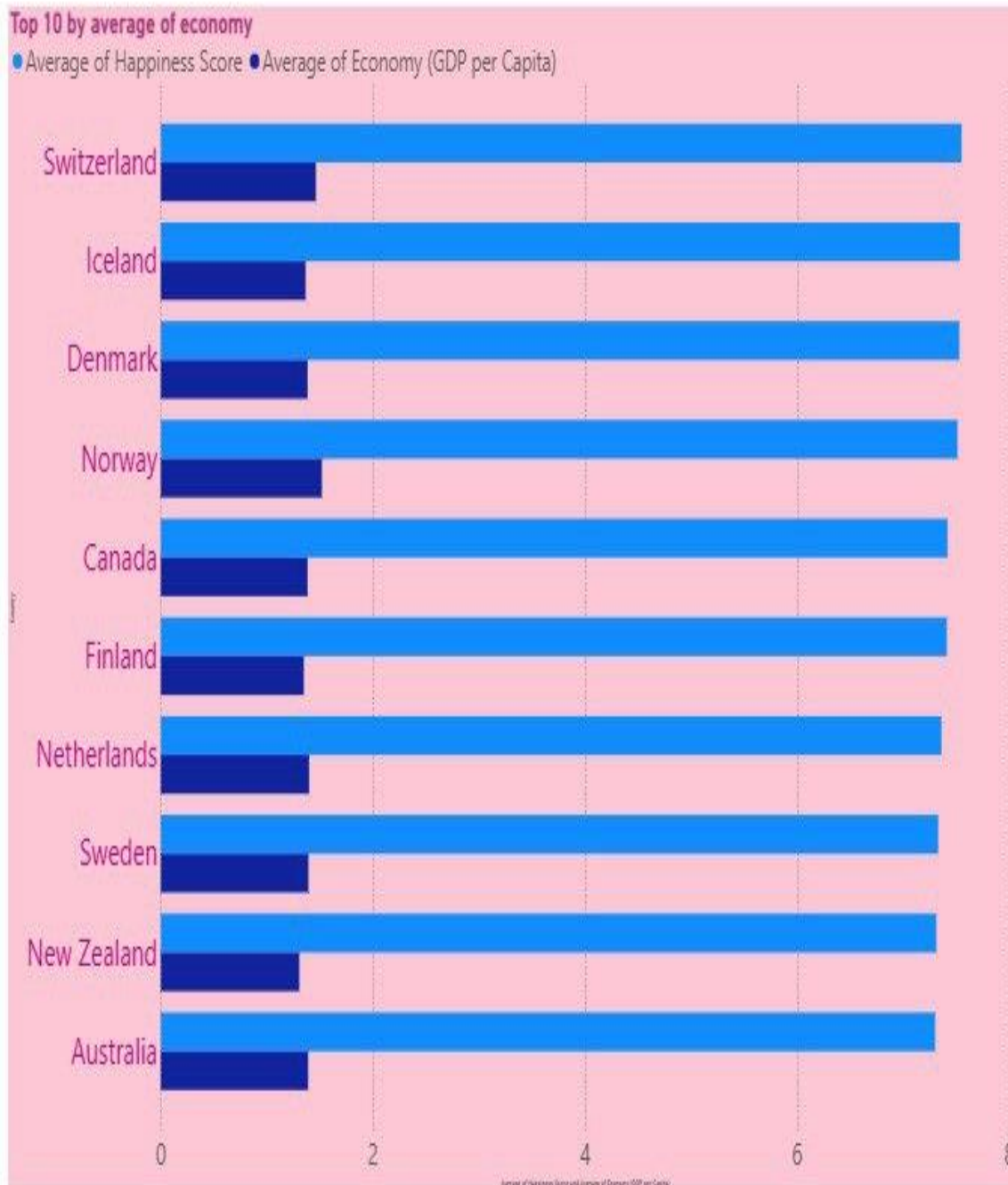
Key Metrics:

Maximum: 7.55 (Highest happiness score observed)

Average: 5.38 (Overall mean happiness)

Minimum: 2.84 (Lowest happiness score observed)

Insight: Reveals a wide happiness range (~4.75 points difference) across countries, with the average (5.38) closer to the midpoint of the scale.



5. Top 10 Countries by Economic Performance

Chart Type: Bar Chart (Horizontal)

Purpose: Ranks countries by their average Economy (GDP per capita) scores.

Interesting Finding: The top 10 countries by economic performance are identical to the top 10 by happiness score, strongly suggesting a correlation between economic prosperity and overall happiness in global rankings.

World Happiness Data Summary

This analysis of global happiness data (2015-2016) reveals key patterns:

Top Performers: Switzerland leads in both happiness (7.60) and economic performance, followed by Nordic and Western European countries. The same 10 nations dominate both happiness and wealth rankings.

Global Range: Happiness scores vary widely from 2.84 (lowest) to 7.55 (highest), with an average of 5.38 globally.

Regional Insights: Sub-Saharan African nations like Sudan show stronger family support than health metrics, highlighting regional disparities in well-being factors.

Yearly Stability: Countries like Bulgaria maintained consistent scores (4.22) between 2015-2016, suggesting relative stability in national happiness.

Core Finding: Economic prosperity strongly correlates with happiness, with wealthy nations consistently reporting higher life satisfaction across all metrics.



Thank you